

Junho Lee 이준호

Medical AI · Full-Stack Development · Undergraduate Researcher

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Education

Jeonbuk National University (JBNU) — B.S. in Computer & Artificial Intelligence

Mar 2023 – Feb 2027 (expected)

- Major GPA 4.4 / 4.5 · Overall 4.25 / 4.5 · Jeonju, South Korea

Research Experience

Macslab (Medical AI Laboratory), JBNU — Undergraduate Researcher

Jun 2025 – Present

- Advisor: Prof. Kyungsu Lee · research on medical imaging AI
- Read and presented deep-learning papers; implemented an MNIST digit-classification model
- Deployed web services and integrated inference pipelines for medical imaging AI projects

Selected Projects

Squat OOD Safety Evaluation · First author · Accepted at KIISE 2026

Python · MC-Dropout

- Safety evaluation that rejects out-of-distribution (OOD) motions via MC-Dropout (BCNN1D) uncertainty in single-RGB-camera squat analysis (Advisor: Prof. Kyungsu Lee)
- Key finding — ID accuracy ≠ OOD detection: top accuracy ConvNeXt1D (90.1%) vs top OOD AUROC BCNN1D (74.8/75.1%), 5-fold vs 4 deterministic models

AFF Diagnosis System · Medical Imaging AI · Full-Stack

Vue3+TS · FastAPI · AWS

- Turned an Atypical Femoral Fracture (AFF) detection model into a clinically usable web diagnostic system
- Owned the **system engineering** — DICOM viewer (Window/Level, mm measurement), FastAPI 3-stage inference serving, AWS deployment (model training provided by the lab)

Knee-Angle Residual Correction · Pose Estimation · in prep for IEEE JBHI

Python · scikit-learn

- Corrects the systematic knee-angle error of monocular 3D pose estimation via residual learning
- Validated across 4 independent datasets (84 subjects) with LOSO/LODO — MAE 13.74° → 8.47° (+38.4%)

Thyroid Supplement CDSS · Clinical Decision Support

FastAPI · Rule Engine · CI

- Deterministic rule engine for safety/evidence judgments; LLM used only for explanation (reproducible, auditable)
- Integrates PubMed & MFDS public data; Docker/Render deployment with GitHub Actions CI

Other · Full-Stack / Systems

- Calendar collaboration suite (web-service design term project, FastAPI+Vue), real-time multiplayer board game (server & front), Agentic CGR (a proposed agent self-correction pattern)

Technical Skills

Languages Python · TypeScript / JavaScript

ML / AI PyTorch · scikit-learn · TensorFlow · OpenCV

Web FastAPI · Flask · Vue 3 · React · React Native · Flutter

Infra Docker · AWS EC2 · Render · GitHub Actions (CI) · Firebase

Domain Medical Imaging (DICOM) · Pose Estimation · CDSS · RAG / LLM

Certifications

- Engineer Information Processing — HRDK (Feb 2024)